

Additional Literature on Neural Networks with Random / Fixed Weights

1 Foundational Random–Feature Theory

- **Ali Rahimi & Benjamin Recht.** *Random Features for Large-Scale Kernel Machines.* *Advances in Neural Information Processing Systems* 20 (2007), 1177–1184.

2 Random–Feature / Random–Weight Solvers for PDEs

- **Chunyang Liao.** *Solving Partial Differential Equations with Random Feature Models.* arXiv:2501.00288 (2024).
- **Maxim Kuvakin, Zhihui Mei & Jingrun Chen.** *Weak Random Feature Method for Solving Partial Differential Equations.* arXiv:2505.00508 (2025).
- **Yifei Sun & Jingrun Chen.** *Two-Level Random Feature Methods for Elliptic Partial Differential Equations over Complex Domains.* *Computer Methods in Applied Mechanics and Engineering* 441 (2025), 117961.
- **Xi'an Li, Jinran Wu, Yujia Huang, Zhe Ding, Xin Tai, Liang Liu & You-Gan Wang.** *Fourier-Feature Induced Physics-Informed Randomized Neural Network Method to Solve the Biharmonic Equation.* *Journal of Computational and Applied Mathematics* 468 (2025), 116635.
- **Gianluca Fabiani, Evangelos Galaris, Lucia Russo & Constantinos Siettos.** *Parasimonious Physics-Informed Random Projection Neural Networks for Initial-Value Problems of ODEs and Index-1 DAEs.* *Chaos* 33 (2023), 043128.

3 Expressivity and Theory of Frozen-Weight Networks

- **Ezekiel Williams, Avery Hee-Woon Ryoo, Thomas Jiralerspong, Alexandre Payeur, Matthew G. Perich, Luca Mazzucato & Guillaume Lajoie.** *Expressivity of Neural Networks with Random Weights and Learned Biases.* arXiv:2407.00957 (2024).
- **Thu Bui, Carola-Bibiane Schönlieb, Bruno Ribeiro, Beatrice Bevilacqua & Moshe Eliasof.** *On the Effectiveness of Random Weights in Graph Neural Networks.* arXiv:2502.00190 (2025).
- **Nicholas H. Nelsen & Andrew M. Stuart.** *Operator Learning Using Random Features: A Tool for Scientific Computing.* arXiv:2408.06526 (2024).

4 Surveys and Broader Overviews

- **Guang-Bin Huang, Dianhui Wang & Yuan Lan.** *Extreme Learning Machines: A Survey.* *International Journal of Machine Learning and Cybernetics* 2(2) (2011), 107–122.
- **Simone Scardapane & Dianhui Wang.** *Randomness in Neural Networks: An Overview.* *WIREs Data Mining and Knowledge Discovery* 7(2) (2017), e1200.